**94.** Satellite having the same orbital period as the period of rotation of the earth about its own axis is known as
(a) polar satellite
(b) stationary satellite
(c) geostationary satellite
(d) INSAT

**95.** A typical black hole is always specified by
(a) a (curvature) singularity
(b) a horizon
(c) either a (curvature) singularity or a horizon
(d) a charge

**96.** The motion of the wheel of a cycle is
(a) rectilinear
(b) translatory and rotatory
(c) rotatory
(d) translatory

## Previous Years' Questions

**97.** In respect of the difference of the gravitational force from electric and magnetic forces, which one of the following statements is true? **2015 I**
(a) Gravitational force is stronger than the other two
(b) Gravitational force is attractive only, whereas the electric and the magnetic forces are attractive as well as repulsive
(c) Gravitational force has a very short range
(d) Gravitational force is a long range force, while the other two are short range forces

**98.** A brick is thrown vertically from an aircraft flying 2 km above the earth. The brick will fall with a **2015 II**
(a) constant speed
(b) constant velocity
(c) constant acceleration
(d) constant speed for sometime, then with constant acceleration as it nears the Earth

**99.** Which one of the following statements is not correct? **2015 II**
(a) Weight of a body is different on different planets
(b) Mass of a body on the earth, on the moon and in empty space is the same
(c) Weightlessness of a body occurs when the gravitational forces acting on it is counter-balanced
(d) Weight and mass of a body are equal at sea level on the surface of the earth

## Properties of Matter

**100.** What is the value of stress in a wire of steel having radius of 2 mm if 10 KN of force is applied on it?
(a) $7.96 \times 10^8$ N/m$^2$
(b) $8.00 \times 10^7$ N/m$^2$
(c) $10 \times 10^7$ N/m$^2$
(d) $10 \times 10^8$ N/m$^2$

**101.** Why the spring is made up of steel in comparison of copper?
(a) Copper is more costly than steel
(b) Copper is more elastic than steel
(c) Steel is more elastic than copper
(d) None of the above

**102.** Match the following columns.

| Column I (Property) | Column II (Example) |
|---|---|
| A. Elastic | 1. Iron |
| B. Plastic | 2. Steel |
| C. Ductile | 3. Quartz |
| D. Perfectly elastic | 4. Paraffin wax |

**Codes**

| | A | B | C | D | | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| (a) | 2 | 4 | 1 | 3 | (b) | 1 | 2 | 3 | 4 |
| (c) | 4 | 3 | 2 | 1 | (d) | 3 | 4 | 2 | 1 |

**103.** Consider the following statements
1. The internal restoring force acting per unit area of cross-section of the deformed body is called stress.
2. The ratio of stress to strain is a constant for a material and is called modulus of elasticity
3. Air is more elastic than water.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) Only 3

**104.** A 50 kg girl wearing high heal shoes balances on a single heal. The heal is circular with a diameter 1.0 cm. The pressure exerted by the heal on the horizontal floor is (g = 9.8 N) kg
(a) $6.2 \times 10^6$ N/m$^2$
(b) $2.6 \times 10^5$ N/m$^2$
(c) $6.2 \times 10^8$ N/m$^2$
(d) $2.6 \times 10^6$ N/m$^2$

**105.** The pressure exerted on the ground by a man is greatest
(a) when he lies down on the ground
(b) when he stands on the toes of one foot
(c) when he stands with both foot flat on the ground
(d) All of the above yield the same pressure

**106.** Consider the following statements
1. Pressure is measured by barometer.
2. Atmospheric pressure is measured by manometer.
3. Atmospheric pressure decrease with altitude.
4. Atmospheric pressure is 700 torr.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) Only 3 (d) 1 and 4

**107.** The relative density of silver is 10.8. If the density of water be $1.0 \times 10^3$ kg/m$^3$ the density of silver will be
(a) $10.8 \times 10^3$ kg/m$^3$ (b) $9 \times 10^2$ kg/m$^3$
(c) $8 \times 10^3$ kg/m$^3$ (d) $5 \times 10^2$ kg/m$^3$

**108.** The apparent weight of a steel sphere immersed in various liquids is measured using a spring balance. The greatest reading is obtained for the liquid
(a) having the smaller density
(b) having the largest density
(c) in which the sphere was submerged deepest
(d) having the greatest volume

**109.** Hydraulic brakes in automatic vehicles are based on
(a) Archimedes' principle
(b) Toricellis equation
(c) Pascal's law
(d) Bernoulli's equation

**110.** If a ship moves from fresh water into sea water, it will
(a) sink completely
(b) sink a little bit
(c) rise a little higher
(d) remain unaffected

**111.** Archimedes' principle is used
1. determining the relative density of a substance.
2. in designing ships and submarines
3. measuring a terminal velocity in air.

Which of the following statements is/are correct?
(a) 1 and 3 (b) 2 and 3
(c) 1 and 2 (d) Only 2

**112.** A needle made of steel floats in water, if it is placed horizontally because of
(a) capillary action
(b) interference
(c) viscosity
(d) surface tension

**113.** If the temperature of a liquid is raised, then its surface tension is
(a) decreased (b) increased
(c) irregular (d) equal to viscosity

**114.** More liquid rises in a thin tube because of
(a) larger value of radius
(b) larger value of surface tension
(c) small value of surface tension
(d) small value of radius

**115.** Coatings used on raincoat are waterproof because
(a) water is absorbed by the coating
(b) cohesive force becomes greater
(c) water is not scattered away by the coating
(d) angle of contact decrease

**116.** Small droplets of a liquid are spherical in shape because of
(a) viscosity (b) surface tension
(c) density (d) conductivity

**117.** A U shaped wire is dipped in a soap solution and removed. The thin soap film formed between the wire and a light slider supports a weight of $1.5 \times 10^{-2}$N. The length of the slider is 30 m. What is the surface tension of the film?
(a) 0.025 N/m (b) 0.052 N/m
(c) 25 N/m (d) None of these

**118.** Due to surface tension
1. Dancing of camphor on water.
2. Floatation of needle on water.

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) Both are wrong

**119.** Cohesive force is experienced between
(a) magnetic substance
(b) molecules of different substances
(c) molecules of same substances
(d) None of the above

**120.** Which one of the following statements is correct?
(a) The angle of contact of water with glass is acute while that of mercury with glass is obtuse
(b) The angle of contact of water with glass is obtuse while that of mercury with glass is acute
(c) Both the angle of contact of water with glass and that of mercury with glass are acute
(d) None of the above

**121.** In an oil lamp, the oil rises up in the wick because
(a) oil is volatile (b) oil is very light
(c) of the capillary action
(d) of the surface tension

**122.** The rise of a liquid in a capillary tube is due to
(a) diffusion (b) osmosis
(c) surface tension (d) viscosity

**123.** The action of a nib spilt at the top is explained by
(a) gravity flow (b) diffusion of fluid
(c) capillary action (d) osmosis of liquid

**124.** Choose the wrong statement among the followings.
(a) Surface tension is a molecular phenomenon
(b) Tiny drops of a liquid are spherical due to spherical tension
(c) Oil rises in the wick due to capillarity
(d) Cold drink rises in the straw due to capillarity

**125.** Due to capillary action
1. the oil in the wick of the lamp rises.
2. refill pen writes

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) 1 and 2
(d) None of the above

**126.** Machine parts get jammed in winter, because
(a) the viscosity of the lubricant used in machine parts increases with the decrease in temperature
(b) the viscosity of lubricant used in machine parts decreases with the decrease in temperature
(c) the inertia of machine parts increases with the decrease in temperature
(d) the inertia of machine parts increases with the decrease in temperature

**127.** After rising a short distance the smooth column of smoke from a cigarette breaks up into an irregular and random pattern. In a similar fashion, a stream of fluid flowing past an obstacle breaks up into eddies and vortices which give the flow irregular velocity components transverse to the flow direction. Other examples include the wakes left in water by moving ships the sound produced by whistling and by wind instruments. These examples are the results of
(a) laminar flow of air
(b) streamline flow of air
(c) turbulent flow of air
(d) viscous flow at low speed

**128.** Consider the following statements.
1. The property of the liquid by virtue of which it opposes the relative motion between its adjacent layers is known as viscosity.
2. Viscosity of liquid increases with increase in pressure.
3. Viscosity of a fluid is measurement by its coefficient of viscosity.
4. Viscosity is not property of gases.

Which of the following statements is/are correct?
(a) 1, 2 and 3 (b) 3 and 4
(c) 2 and 4 (d) Only 4

**129.** Hair of a shaving brush cling together when the brush is removed from water due to
(a) viscosity (b) surface tension
(c) friction (d) elasticity

**130.** Match the following columns.

| Column I | Column II |
|---|---|
| A. Hook's law | 1. Lactometers |
| B. Pascal Law | 2. Elasticity |
| C. Archimedes' Principle | 3. Hydraulic press |
| D. Stokes' law | 4. Terminal velocity |

**Codes**

| | A | B | C | D | | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| (a) | 2 | 3 | 1 | 4 | (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 2 | 1 | 4 | (d) | 1 | 2 | 3 | 4 |

## Previous Years' Questions

**131.** A liquid is kept in a regular cylindrical vessel upto a certain height. If this vessel is replaced by another cylindrical vessel having half the area of cross-section of the bottom, the pressure on the bottom will **2012 I**
(a) remain unaffected
(b) be reduced to half the earlier pressure
(c) be increased to twice the earlier pressure
(d) be reduced to one-fourth the earlier pressure

**132.** It is difficult to cut things with a blunt knife because **2012 II**
(a) the pressure exerted by knife for a given force increases with increase in bluntness
(b) a sharp edge decreases the pressure exerted by knife for a given force

(c) a blunt knife decreases the pressure for a given force
(d) a blunt knife decreases the area of intersection

**133.** When deep sea fishes are brought to the surface of the sea, their bodies burst. This is because the blood in their bodies flows at very **2012 II**
(a) high speed
(b) high pressure
(c) low speed
(d) low pressure

**134.** Which type/types of pen uses/use capillary action in addition to gravity for flow of ink? **2013 I**
(a) Fountain pen
(b) Ballpoint pen
(c) Gel pen
(d) Both ballpoint and gel pens

**135.** After rising a short distance the smooth column of smoke from a cigarette breaks up into an irregular and random pattern. In a similar fashion, a stream of fluid flowing fast an obstacle breaks up into eddies and vortices which give the flow irregular velocity components transverse to the flow direction. Other examples include the wakes left in water by moving ships the sound produced by whistling and by wind instruments. These examples are the results of **2013 II**
(a) laminar flow of air
(b) streamline flow of air
(c) turbulent flow of air
(d) viscous flow at low speed

**136.** Dirty cloths containing grease and oil stains are cleaned by adding detergents to water. Stains are removed because detergent **2013 II**
(a) reduces drastically the surface tension between water and oil
(b) increases the surface tension between water and oil
(c) increases the viscosity of water and oil
(d) decreases the viscosity of detergent mixed water

## Heat

**137.** The value of – 30°C on Fahrenheit scale is
(a) –54° F (b) 32° F
(c) 22° F (d) –22° F

**138.** Consider the following statements
1. Fahrenheit is the smallest unit measuring temperature.
2. Fahrenheit was the first temperature scale used for measuring temperature.

Which of the following statements is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) None of the above

**139.** The temperature for which the reading on Celsius and Fahrenheit scales are identical, is
(a) – 40° C (b) 40° C
(c) – 30° C (d) 30° C

**140.** Temperature of the body measures
(a) hotness of the body
(b) amount of heat contained in the body
(c) degree of hotness of the body
(d) the average energy of molecules of the object

**141.** The instrument used to mark the highest fixed point of a thermometer is
(a) pyrometer (b) hydrometer
(c) spherometer (d) calorimeter

**142.** *P* and *Q* are two objects. The temperature of *P* is greater than that of *Q*. This means that
(a) the heat content of *P* will always be greater than that of *Q*
(b) the average potential energy of *P* is greater than the average potential energy of *Q*
(c) the total energy of *P* is greater than the total energy of the molecules of *Q*
(d) the molecules of *P* are moving faster on the average than the molecules of *Q*

**143.** Which one of the following statements is true?
(a) Temperatures differing 25° on the Fahrenheit (F) scale must differ by 45° on the celsius (C) scale
(b) 0°F corresponds to – 32° C
(c) Temperatures which differ by 10° on the celsius scale must differ by 18° on the Fahrenheit scale
(d) Water at 90°C is warmer than water at 202°F.

**144.** Consider the following statements
1. Mercury is used in clinical thermometer for measuring body temperature.
2. Mercury shines and is easily observable.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

**145.** Specific heat of a substance depends upon
(a) mass of the substance
(b) volume of the substance
(c) shape of the body
(d) nature of the substance

**146.** Vapour pressure of liquid depends upon the
(a) quantity of heat
(b) temperature of liquid
(c) specific heat of liquid
(d) density of liquid

**147.** Two bodies *A* and *B* are of same mass and same amount of heat is given to both of them. If the temperature of *A* decreases more than that of *B* because of heat addition, then
(a) the specific heat capacity of *A* is more than that of *B*
(b) the specific heat capacity of *A* is less than that of *B*
(c) both *A* and *B* have the same specific heat capacity but *A* has greater thermal conductivity
(d) both *A* and *B* have the same specific heat capacity but *B* has greater thermal conductivity

**148.** Consider the following statements
1. Specific heat is amount of heat required to raise the temperature of a unit of the substance by 1°C.
2. The specific heat of water is maximum
3. Gases have no specific heat.

Which of the following statements is/are correct?
(a) 1 and 2 (b) 2 and 3
(c) 2 and 1 (d) 1 and 3

**149.** Consider the following statements
1. Cooking utensils have low specific heat and high conductivity.
2. Specific heat is the amount of heat required to raise the temperature of unit mass of mass of the substance by 1°C

Which of the following statements is/are correct?
(a) 1 and 2 (b) Only 1
(c) Only 2 (d) None of these

**150.** Real expansion of a liquid is always
(a) greater than apparent expansion
(b) less than apparent expansion
(c) equal to apparent expansion
(d) None of the above

**151.** Match the following columns.

| Column I | Column II |
|---|---|
| A. Bimetallic strip | 1. Radiation from a hot body |
| B. Steam engine | 2. Energy conversion |
| C. Incondescent lamp | 3. Melting |
| D. Electric fuse | 4. Thermal expansion of solids |

Codes

| | A | B | C | D | | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| (a) | 4 | 2 | 1 | 3 | (b) | 3 | 2 | 1 | 4 |
| (c) | 4 | 3 | 2 | 1 | (d) | 2 | 1 | 3 | 4 |

**152.** Match the following columns.

| Column I | Column II |
|---|---|
| A. Freezing point of mercury | 1. 0°C |
| B. Freezing point of ice | 2. – 115°C |
| C. Freezing point of alcohol | 3. – 39°C |

Codes

| | A | B | C | | A | B | C |
|---|---|---|---|---|---|---|---|
| (a) | 1 | 2 | 3 | (b) | 2 | 3 | 1 |
| (c) | 3 | 1 | 2 | (d) | 1 | 3 | 2 |

**153.** Latent heat of fusion of any body
(a) is independent of pressure
(b) is dependent of volume
(c) is always same
(d) varies under different conditions

**154.** Water is used in hot water bottles in preference to any other liquid because
(a) water has greatest latent heat
(b) water is good conductor of heat
(c) water has greatest specific heat
(d) water has high boiling point

**155.** Consider the following statements
1. Steam is more harmful for human body than the boiling water in case of burn.
2. Boiling water contains more heat than steam.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

**156.** Factors affecting evaporation of a liquid are
1. the nature of the liquid
2. temperature of surrounding
3. humidity of surrounding air
4. temperature of liquid

Which of the following statements is/are correct?
(a) Only 2
(b) Only 3
(c) 1, 2 and 3
(d) All of the above

**157.** Consider the following statements
1. Water kept in an open vessel will quickly evaporate on the surface of the moon.
2. The temperature at the surface of the moon is much higher than the boiling point of the water.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

**158.** Heat energy of an object is
(a) the average energy of the molecules of the object
(b) the total energy of the molecules of the object
(c) the average velocity of the molecules of the object
(d) the average potential energy of the molecules of the object

**159.** A glowing electric bulb becomes hot after some time because the heat from filament is transmitted to the glass bulb by
(a) conduction (b) convection
(c) radiation (d) another process

**160.** Diesel engine
1. works with a oil plug.
2. is associated with the risk of explosion.
3. has larger efficiency (58%)

Which of the following statements is/are correct?
(a) Only 1
(b) 2 and 3
(c) Only 2
(d) All of the above

**161.** If there were no atmosphere, the average temperature on the surface of the earth would be
(a) higher (b) lower
(c) same as today (d) 0° C

**162.** Transfer of heat energy from the sun to the moon takes place by
(a) radiation only
(b) radiation and conduction
(c) radiation and convection
(d) radiation, conduction and convection

**163.** Material used for making cooking utensils should have
(a) low specific heat and high thermal conductivity
(b) high specific heat and high thermal conductivity
(c) low specific heat and low thermal conductivity
(d) high specific heat and low thermal conductivity

**164.** The wet clothes dry more quickly on a warm day because
(a) air carries less moisture
(b) air can absorb moisture rather quickly
(c) high temperature of the air helps fast evaporation
(d) All of the above

**165.** A glass chimney stops an oil lamp from smoking because
(a) it increases the supply of oxygen to the flame by connection
(b) the heat produced ensures complete combustion of carbon particles
(c) Both 'a' and 'b'
(d) None of the above

**166.** Consider the following statements
1. The water in a vessel placed on fire becomes hot by convection.
2. Heat is also transmitted in the form of electromagnetic waves from a hot body.

Which of the following statements is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) None of these

**167.** A perfect black body has the unique characteristic feature as
(a) a good absorber only
(b) a good radiator only
(c) a good absorber and a good radiator
(d) Neither a radiator nor an absorber

**168.** Nights are cooler in the deserts than in the plains because
(a) sand radiates heat more quickly than the earth
(b) the sky remains clear most of the time
(c) sand absorbs heat more quickly than the earth
(d) None of the above

**169.** We are asked to wear white clothes in summer because
(a) they look more graceful
(b) it is a convention
(c) they are visible from long distance
(d) None of the above

**170.** Half portion of a rectangular piece of ice is wrapped with a white piece of cloth while the other half with a black one. In this context, which one among the following statements is correct?
(a) Ice melts more easily under black wrap
(b) Ice melts more easily under white wrap
(c) No ice melts at all under the black wrap
(d) No ice melts at all under the white wrap

**171.** Mr. X was advised by an architect to make outer walls of his house with bricks. The correct reason is that such walls
(a) make the building stronger
(b) help keeping inside cooler in summer and warmer in winter
(c) prevent seepage of mositure from outside
(d) protect the building from lightning

**172.** The blackboard seems black because it
(a) reflects every colour
(b) does not reflect any colour
(c) absorbs black colour
(d) reflects black colour

**173.** Consider the following statements
1. Rough surface is a better radiator than smooth surface.
2. Highly polished surface is a very good radiator.
3. Black surface is a better absorber of radiation than a white one.
4. Black surface is a better radiator than white one.

Which of the following statements is/are correct?
(a) 1, 2 and 3 (b) 1 and 2
(c) 3 and 4 (d) 1, 3 and 4

**174.** A polished plate with a rough black spot is heated to a high temperature and taken in a dark room, then,
(a) spot will appear brighter than the plate
(b) spot will appear darker than the plate
(c) both will appear equally bright
(d) Neither the spot, nor the plate will be visible

**175.** The temperature of the stars can be estimated by
(a) Wien's displacement law
(b) Rayleigh-Jeans law
(c) Faraday's law
(d) Maxwell-Boltzmann law

**176.** A solid copper sphere of radius $R$ and hollow sphere of the same material and outer radius $R$ are heated to the same temperature and allowed to cool in the same environment, the
(a) hollow sphere cools faster
(b) solid sphere cools faster
(c) both spheres cool at the same rate
(d) in the beginning solid sphere cools faster, at the end hollow sphere cools faster

**177.** If the door of a running refrigerator in a closed room is kept open, what will be the net effect on the room?
(a) It will cool the room
(b) It will heat the room
(c) It will make no difference on the average
(d) It will make thte temperature go up and down

**178.** The gas used in a refrigerator is
(a) cooled down on flowing
(b) heated up on flowing
(c) cooled down, when compressed
(d) cooled down, when expanded

## Previous Years' Questions

**179.** The celsius temperature is a/an
(a) relative temperature **2013 I**
(b) absolute temperature
(c) specific temperature
(d) approximate temperature

**180.** The gas used in a refrigerator is
(a) cooled down on flowing **2013 II**
(b) heated up on flowing
(c) cooled down when compressed
(d) cooled down when expanded

**181.** A ray of white light strikes the surface of an object. If all the colours are reflected the surface would appear **2013 II**
(a) black (b) white (c) grey (d) opaque

**182.** Two layers of a cloth of equal thickness provide warmer covering than a single layer of cloth with double the thickness. Why? **2014 I**
(a) Because of the air encapsulated between two layers
(b) Since effective thickness of two layers is more
(c) Fabric of the cloth plays the role
(d) Weaving of the cloth plays the role

**183.** Which one of the following physical quantities is the same for molecules of all gases at a given temperature? **2015 II**
(a) Speed (b) Mass
(c) Kinetic energy (d) Momentum

**184.** Two systems are said to be in thermal equilibrium, if and only if **2016 I**
(a) there can be a heat flow between them even if they are at different temperatures
(b) there cannot be a heat flow between them even if they are at different temperatures
(c) there is no heat flow between them
(d) their temperatures are slightly different

## Oscillations and Waves

**185.** The force acting on a particle executing simple harmonic motion is
(a) directly proportional to the displacement and is directed away from the mean position
(b) inversely proportional to the displacement and is directed towards the mean position
(c) directly proportional to the displacement and is directed towards the mean position
(d) inversely proportional to the displacement and is directed away from the mean position

**186.** Three pendulum clocks $A$, $B$ and $C$ are synchronised on the surface of the earth at the sea-level. The clock $A$ is taken to the top of a mountain and the clock $B$ is taken under a deep mine. The temperature at the three locations are assumed to be the same. On the basis of the above, which one of the following is correct?
(a) Both $A$ and $B$ run faster than $C$
(b) Both $A$ and $B$ run slower than $C$
(c) $A$ runs slower than $B$ but runs faster than $C$
(d) $B$ runs slower than $C$ but $A$ runs faster than C

**187.** The length of a simple pendulum becomes four times the original length. If the initial time period of pendulum is $T$, then the new time period is
(a) $T$ (b) $2T$ (c) $3T$ (d) $4T$

**188.** If the amplitude of a simple pendulum is made one-fourth, its time period will become
(a) twice (b) half
(c) one-fourth (d) will not change

**189.** Water waves are
(a) transverse (b) longitudinal
(c) partly transverse, partly longitudinal
(d) sometimes longitudinal, sometimes transverse

**190.** Energy is not transferred by
(a) transverse progressive waves
(b) longitudinal progressive waves
(c) stationary waves
(d) electromagnetic waves

**191.** Waves inside a gas are
(a) longitudinal (b) transverse
(c) partly longitudinal and party transverse
(d) Neither longitudinal nor transverse